Colter Snyder

Lakewood, CO · (720) 626-2558 · coltersnyder@gmail.com · in/colter-snyder-a36854183

SUMMARY

Software engineer who builds web front ends (Angular, TypeScript, HTML/CSS/JavaScript) and the Python and Rust services behind them, and who has worked on client-side web projects since 2021. Builds tools that turn performance data into charts people can act on, uses AI-assisted development tools (Claude Code, Cursor) within strict data limits, and has taught more than 200 students. M.S. in Computer Science (3.97 GPA); professional working Spanish.

EXPERIENCE

Innovative Scientific Solutions, Inc.

Software Engineer II — *Dec 2024 – Present*

- Develop web user interfaces in Angular, TypeScript, and Tauri for probabilistic nuclear weapons effects simulation platforms, and contribute extensively to UI/UX design.
- Build Python (Flask) and Rust back-end services with WebSockets and MongoDB, deployed with Docker on Linux.
- Spearheaded an internal Python performance-benchmarking tool (a CLI, then a GUI front end) that runs any combination of the platform's 13 effect calculations and charts the results; it exposed a 10x slowdown in our interface to an external library, made the case to leadership with clear graphs, and saved an estimated several weeks of manual testing with hand-crafted JSON inputs.
- Take part in technical meetings with government customers, and lead development of individual features from requirements and design documentation through implementation and automated testing; review teammates' merge requests on every change.
- Apply AI-assisted tools (Claude Code, Cursor) only to non-controlled exploration and research, keeping all ITAR- and CUI-controlled code and data out of commercial AI services.

Tyler Technologies — Lakewood, CO

Software Development Intern — *May 2023 – Aug 2023*

- Diagnosed and repaired a 2,000-test automated suite, restoring about 1,500 tests (75%) to passing.
- Migrated a major legacy VB6 desktop application to C# as part of a cloud modernization effort.

Colorado School of Mines — Golden, CO

Postgraduate Research Assistant, Department of Computer Science — *May 2024 – Dec 2024*

- Developed object detection (machine learning) models for semiconductor cleanroom equipment and integrated them, with Python back-end services, into a Unity AR training app with the University of Arizona.

Graduate Research & Teaching Assistant — *Jan 2024 – May 2024*

- Researched graph databases for water treatment plant operations on a National Alliance for Water Innovation (NAWI) project (advisor: Dr. Gabe Fierro).

Computer Science Outreach Webmaster — *Aug 2021 – May 2022*

- Developed and maintained the DECtech outreach website (tech.mines.edu) in HTML, CSS, and JavaScript.

TEACHING & KNOWLEDGE SHARING

Colorado School of Mines — Golden, CO

Adjunct Professor, Department of Computer Science — *Jan 2023 – Dec 2023*

- Co-taught CSCI 220: Data Structures and Algorithms (C++) to 120 undergraduates and taught CSCI 102 (Python) to 90, troubleshooting student code and explaining technical concepts to varied audiences.
- Designed and taught STEM summer camps for grades 3–10 and co-presented a teacher professional development workshop at InnEdCO 2021.

SKILLS

Web: Angular, TypeScript, JavaScript, HTML/CSS, HTTP & WebSockets, Tauri, UI/UX Design

Languages: Python, Rust, C++, C#, SQL, Fortran

Data, Cloud & Systems: MongoDB, SQL, Graph Databases, Docker, Linux, Git, GitLab CI/CD, SonarQube

AI & ML: Claude Code, Cursor, Object Detection / Computer Vision

Spoken Languages: English (native), Spanish (professional working), German (limited working)

EDUCATION

Colorado School of Mines — Golden, CO

Master of Science in Computer Science • GPA 3.97 — *Jan 2022 – May 2024*

Bachelor of Science in Computer Science • Magna Cum Laude • GPA 3.77 — *Aug 2019 – Dec 2022*

Relevant coursework: Computer Vision, Information Security & Privacy, IoT Security & Privacy, Theory of Cryptography

Honors: CACI C-MAPP Scholar, National Merit Scholar, Dean's List, Eagle Scout, Rotary Youth Leadership Award